

Solution Requirements

Digitize Metro Ticket Booking Using ServiceNow

The solution digitizes the metro ticket booking process and replaces queue-based or manual interactions with a guided ServiceNow workflow.

Functional Requirements

ID	Requirement	Description
FR-01	Passenger Booking Form	Allow passengers to enter source station, destination, journey date and number of tickets.
FR-02	Input Validation	Validate mandatory fields and prevent invalid source/destination combinations.
FR-03	Fare Calculation	Calculate the fare automatically based on route, distance or configured fare rules.
FR-04	Booking Creation	Create a unique booking record with passenger and journey information.
FR-05	QR Ticket	Generate an instantly accessible QR-code digital ticket linked to the booking.
FR-06	Notifications	Send booking confirmation and ticket information through automated notifications.
FR-07	Booking Status	Allow users or support staff to view booking status such as Confirmed, Used or Cancelled.
FR-08	Cancellation	Support cancellation according to configured business rules.
FR-09	Admin Management	Allow authorized users to manage stations, fare rules and booking records.
FR-10	Reporting	Provide dashboards for bookings, routes, demand trends and failed transactions.

Non-Functional Requirements

ID	Requirement	Expected Outcome
NFR-01	Performance	Booking and confirmation workflow should respond quickly under normal load.
NFR-02	Availability	The booking service should be available to passengers with minimal downtime.
NFR-03	Security	Use authentication, roles and ACLs to protect booking and passenger data.
NFR-04	Usability	Provide a simple, guided form that can be completed with minimal steps.
NFR-05	Scalability	Support increasing stations, routes and booking volumes without redesigning the core process.
NFR-06	Auditability	Maintain traceable booking and workflow records for operational review.

Business Rules

1. Source and destination stations are mandatory. 2. Source and destination cannot be identical. 3. Journey date must be valid. 4. Fare is calculated before final confirmation. 5. Each confirmed booking receives a unique booking reference and QR token. 6. Only authorized roles can modify fare rules or administrative data.

Acceptance Criteria

A passenger can submit a valid booking, receive an automatically calculated fare, obtain a unique digital QR ticket, and receive confirmation without manual intervention. Administrators can view booking records and operational reports.